A One-Point Star Rigidification for Bounded Extreme-Free Spaces

Run `fa_banach_001`, lane 18

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Source And Open Problem

The source paper is Cuth–Doucha–Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266 [1]. In Section 8.2 it asks:

> 8.2. **Problems.** We find the following question the most pressing.
>
> **Question 1.** Does there exist a Lipschitz-free rigid metric space $\mathcal{M}$ such that $\bigcup E_{ext}(\mathcal{M})$ is not dense in $\mathcal{M}$?
>
> A very interesting answer to the previous question would be answering positively the next question. However, even a negative answer to the next question would be interesting.
>
> **Question 2.** Is $\mathbb{R}^d$, for $d \geqslant 2$, Lipschitz-free rigid? Either if it is equipped with the euclidean or with the Manhattan distance.
>
> **Question 3.** Does every metric space isometrically embed into a Lipschitz-free rigid space that contains only one additional point?
>
> References

We prove another infinite subcase of Question 3, complementary to the finite one-point result from this run.

Proof Intuition

If M is bounded and has no preserved-extreme old pairs, then adding a single point p at a common distance $R > \mathrm{diam}(M)$ creates a pure preserved-extreme star. The source paper's isometry correspondence says that every linear isometry of the free space is encoded by a bijection of the directed star edges satisfying a norm identity for every star path.

A star-edge bijection can permute the leaves and can, a priori, reverse each spoke independently. But the path $x \to p \to y$ has length $d(x, y) < R$. If the two image spokes have opposite orientations, the corresponding molecule has to transport one unit from each of two leaves to p, so its norm is $2R$, impossible. Thus the orientation sign is global. Once the sign is global, the same path identity becomes simply $d(x, y) = d(f(x), f(y))$, so the leaf permutation is an isometry of M.

Result

Theorem 1 (Bounded extreme-free one-point star rigidification)**.** *Let M be a bounded metric space such that $E_{\mathrm{ext}}[M] = \emptyset$. Choose $R > \operatorname{diam}(M)$, add a point $p \notin M$, and define*

$$d(p, x) = R, \qquad x \in M,$$

leaving the metric on M unchanged. Then $M^+ = M \cup \{p\}$ is Lipschitz-free rigid.

Proof. The displayed formula defines a metric: for $x, y \in M$,

$$|d(p, x) - d(p, y)| = 0 \leq d(x, y) \quad \text{and} \quad d(x, y) \leq \operatorname{diam}(M) < 2R = d(x, p) + d(p, y).$$

We first identify the preserved-extreme graph. Since $E_{\mathrm{ext}}[M] = \emptyset$, no old pair $\{x, y\} \subset M$ is preserved extreme in M. Adding p cannot turn such a pair into a preserved extreme pair, because the same witnesses or approximating metric segments inside M remain available in the larger space.

Every radial pair $\{p, x\}$, however, is preserved extreme in M^+. By the source paper's preserved-extreme criterion [1, Fact 2.1], it is enough to show a positive gap away from the endpoints. Fix $\varepsilon > 0$. If $z \in M^+$ satisfies

$$\min\{d(p, z), d(x, z)\} \geq \varepsilon,$$

then $z \in M \setminus \{x\}$, and

$$d(p, z) + d(z, x) - d(p, x) = R + d(z, x) - R = d(z, x) \geq \varepsilon.$$

Thus the radial molecule $m_{p,x}$ is preserved extreme. Consequently

$$E_{\mathrm{ext}}[M^+] = \{(p, x), (x, p) : x \in M\}.$$

This directed star is weakly admissible: it is symmetric, connected, and its vertex set is all of M^+.

Let $T : \mathcal{F}(M^+) \to \mathcal{F}(M^+)$ be a surjective linear isometry. By the source paper's weak Prague correspondence [1, Theorem 3.4], T induces a symmetric bijection $\sigma : E_{\mathrm{ext}}[M^+] \to E_{\mathrm{ext}}[M^+]$ satisfying the source conditions (SA), (SB), and (SC), and

$$T(m_e) = m_{\sigma(e)}, \qquad e \in E_{\mathrm{ext}}[M^+].$$

Since the graph is a star, there are a bijection $f : M \to M$ and signs $\eta_x \in \{\pm 1\}$ such that

$$\sigma(p, x) = \begin{cases} (p, f(x)), & \eta_x = 1, \\ (f(x), p), & \eta_x = -1. \end{cases}$$

We claim the signs are constant. Take distinct $x, y \in M$. Apply condition (SC) to the directed path $(x, p), (p, y)$ from x to y. Since both source edges have length R, it gives

$$d(x, y) = \left\| R\, m_{\sigma(x,p)} + R\, m_{\sigma(p,y)} \right\|.$$

If $\eta_x \neq \eta_y$, the vector on the right is either

$$\delta_{f(x)} + \delta_{f(y)} - 2\delta_p \quad \text{or} \quad 2\delta_p - \delta_{f(x)} - \delta_{f(y)}.$$

Its Arens–Eells norm is $2R$: the only negative mass is at p, or the only positive mass is at p, and each of the two unit masses must travel distance R. Hence $d(x, y) = 2R$, contradicting $d(x, y) \leq \operatorname{diam}(M) < R$. Thus all η_x's are equal to a single sign η.

With this global sign, the same condition (SC) gives, for all $x, y \in M$,

$$d(x,y) = \left\|\delta_{f(x)} - \delta_{f(y)}\right\| = d(f(x), f(y)).$$

Therefore f is a surjective isometry of M. Extending f by $f(p) = p$ gives a surjective isometry of M^+, and T is exactly the induced molecule map, multiplied by the global sign η. This is precisely Lipschitz-free rigidity. $\square$

Remark 1. *Every bounded geodesic metric space satisfies the hypothesis $E_{\mathrm{ext}}[M] = \emptyset$, because between any two distinct points there is a non-endpoint metric midpoint. Thus Theorem 1 gives one-point rigidifications for all bounded geodesic spaces.*

Verification And Limitations

This is a genuine infinite subcase of the source question. It does not settle the full problem. The remaining difficulty is an infinite metric space with old preserved-extreme pairs: then the one-point extension has both radial and old preserved-extreme edges, and a full proof must rule out admissible bijections that mix the two edge types.

Novelty check: local run indexes contained only the source paper and the earlier finite packet from this run. Web searches on 2026-07-03 for "Lipschitz-free rigid one additional point", "one-point Lipschitz-free rigid metric space", and the arXiv id 2402.08266 found the source paper but no later settlement.

The scratch script `code/constant_cone_search.py` checks small finite constant-cone examples. It is included only as exploratory evidence; the proof above is independent.

Human Review Recommendation

Reviewers should check the claim that no old pair becomes preserved extreme after adding p, and the transport-norm computation giving norm $2R$ when two star spokes have opposite image orientations.

References

[1] M. Cuth, M. Doucha, and T. Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266, 2024. https://arxiv.org/abs/2402.08266